

$$\begin{aligned}
 &\equiv (p \wedge q) \vee (q \vee p) \quad (1st \text{ Commutative Law}) \\
 &\equiv (q \vee p) \vee (p \wedge q) \quad (1st \text{ Commutative Law}) \\
 &\equiv q \vee (p \vee (p \wedge q)) \quad (1st \text{ Associative Law}) \\
 &\equiv q \vee ((p \wedge T) \vee (p \wedge q)) \quad (1st \text{ Identity Law}) \\
 &\equiv q \vee (p \wedge (T \vee q)) \quad (2nd \text{ distributive Law}) \\
 &\equiv q \vee (p \wedge (q \vee T)) \quad (1st \text{ commutative Law}) \\
 &\equiv q \vee (p \wedge T) \quad (1st \text{ Domination Law}) \\
 &\equiv q \vee p \quad (1st \text{ identity Law})
 \end{aligned}$$

11) Show that each of these conditional statements is a tautology by using

truth tables

a) $(p \wedge q) \rightarrow p$

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

b) $p \rightarrow (p \vee q)$

p	q	$p \vee q$	$p \rightarrow (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

c) $\neg p \rightarrow (p \rightarrow \neg q)$

p	q	$p \rightarrow \neg q$	$\neg p$	$\neg p \rightarrow (p \rightarrow \neg q)$
T	T	F	F	T
T	F	T	F	T
F	T	F	T	T
F	F	T	T	T

d) $(p \wedge q) \rightarrow (p \rightarrow \neg q)$

p	q	$p \wedge q$	$p \rightarrow \neg q$	$(p \wedge q) \rightarrow (p \rightarrow \neg q)$
T	T	T	F	F
T	F	F	T	T
F	T	F	F	T
F	F	F	T	T

e) $\neg(p \rightarrow \neg q) \rightarrow p$

p	q	$p \rightarrow \neg q$	$\neg(p \rightarrow \neg q)$	$\neg(p \rightarrow \neg q) \rightarrow p$
T	T	F	T	T
T	F	T	F	T
F	T	F	T	F
F	F	T	F	T